

CARNEGIE MELLON UNIVERSITY / EXECONLINE AGENTIC AI PROGRAM

DragonWings Power Availability Profile
Forecaster Agent

Final Capstone Report - Section B

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Abstract

The DragonWings Power Availability Profile (PAP) Forecaster addresses a practical operating question: how much additional electrical load can a portable solar-and-battery system support over the next 12-24 hours while preserving reserve? This report consolidates six capstone checkpoints into a bounded, read-only decision-support architecture. Structured tools establish current state; numerical services forecast, calculate, and validate availability; validated retrieval supplies external memory; and two specialized language-model agents examine ambiguous cases under LangGraph control. The design evolved from a single coordinating agent to selective generation and critique with explicit stopping rules and risk-based human review. The reported evidence consists of a synthetic retrieval fixture and a branch-recovery trace; aggregate accuracy, calibration, latency, and safety rates remain unmeasured in the supplied checkpoint record.

1. Problem and Intended User

Battery state of charge alone does not establish future usable power. Solar production, weather uncertainty, site demand, equipment status, operating limits, and reserve requirements interact across time. An apparently feasible load now may consume energy needed later. Operators need an interval-based view of additional availability, its evidence, and the conditions under which it should be trusted. [C1]

The intended users are energy-system operators, engineers, and facility managers deciding whether to connect optional load, share energy, or preserve capacity. A useful answer combines a machine-readable profile with a concise explanation: what is available, for which interval, which limits apply, how fresh the inputs are, and whether further review is required. The operator remains responsible for any subsequent equipment action.

2. Goal and Scope

The goal is a timely, explainable PAP for a 12-24 hour horizon, with interval power, confidence or uncertainty, freshness, provenance, and constraints. Success requires both useful forecasts and a dependable refusal to publish when current evidence or hard validation is inadequate. Numerical feasibility is evaluated over the full horizon so the same battery energy is not repeatedly allocated to different intervals. [C1,C2]

In scope	System boundary
Telemetry and weather ingestion; PV/demand forecasting; reserve-aware calculation; validated memory; operator guidance and evaluation.	Public, synthetic, or anonymized course data. Customer identities, locations, device identifiers, and proprietary details are excluded.
Automatic publication of supported, valid read-only profiles; bounded recovery; selective human review.	No commands to batteries, inverters, generators, or loads. Extension to other power profiles and physical control is outside this capstone.

3. Final Architecture and Major Components

The final design separates workflow control, probabilistic interpretation, and numerical authority. LangGraph StateGraph owns shared state, routing, persistence, and termination. LangChain create_agent is confined to the Thought Generator and Critic nodes on the ambiguous path. MCP provides bounded read-only access to external resources and tools; it does not manage branch state. [C4-C6; 7-9]

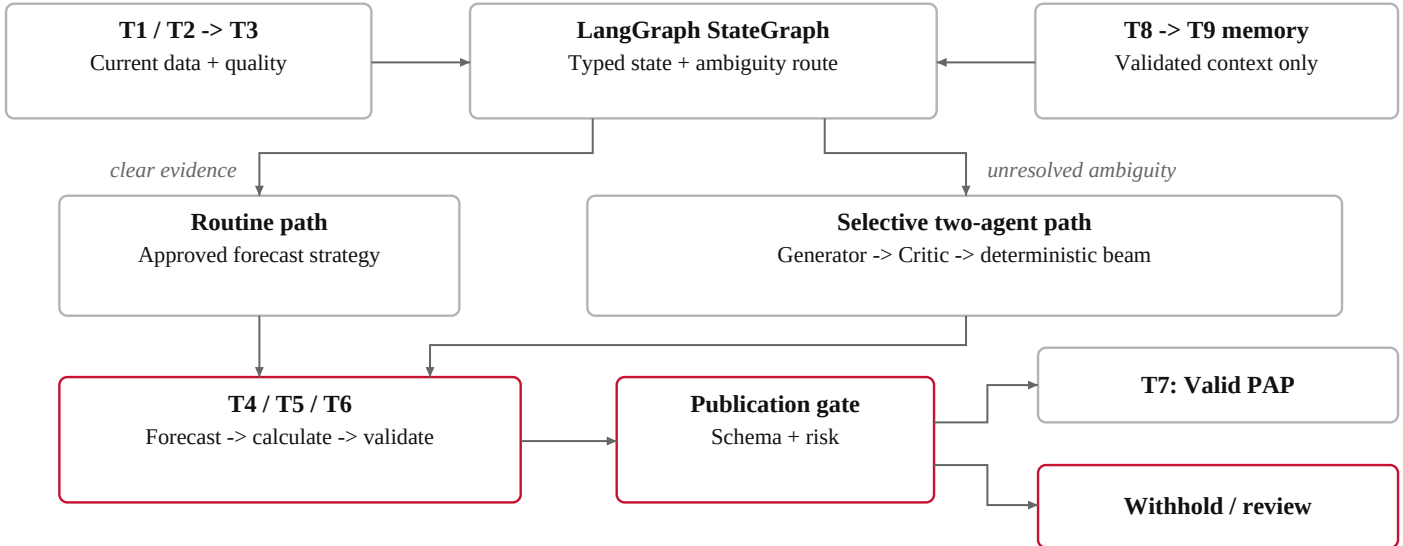


Figure 1. Final logical architecture. Both paths converge on T4/T5/T6. Red outlines identify numerical validation or withholding boundaries. Published forecasts and later outcomes return to T8 for evaluation; only admitted records become eligible for T9.

ID / component	Responsibility and authority
T1 Telemetry adapter	Read anonymized SOC, PV generation, load, alarms, and equipment status, with timestamps and source identifiers.
T2 Weather / solar adapter	Read public cloud, irradiance, wind, forecast uncertainty, and source metadata.
T3 Quality / normalization	Check required fields, freshness, timestamp coherence, units, ranges, and cross-source consistency before reasoning.
T4 PV / demand forecaster	Produce interval generation and demand estimates using an approved numerical model or documented fallback.
T5 Availability calculator	Compute additional interval power and energy deterministically, accounting for existing demand, reserve, and equipment capability.
T6 Constraint validator	Reject infeasible profiles, reserve violations, invalid equipment states, stale evidence, and broken output invariants.
T7 PAP publisher	Emit a typed profile and operator explanation only after numerical, evidence, schema, and risk gates pass.
T8 Outcome / memory evaluator	Compare forecasts with outcomes, track calibration, and admit, reject, or expire memory records.
T9 Semantic retrieval adapter	Retrieve relevant validated history or guidance with configuration, validity, relevance, and provenance filters.

T1-T8 originate in the modular tool design; T9 is the bounded semantic-memory addition. These nine services are not nine agents. Routine cases bypass the two LLM agents and follow the validated numerical path. No model score, memory record, or human acceptance of prose bypasses T5/T6. [C2-C6]

3.1 Retrieval and Memory

Semantic retrieval is reserved for contextual knowledge: validated forecast-outcome episodes, synthetic operating guidance, and public references. Live telemetry, weather measurements, reserve policy, equipment limits, and calculations remain structured authoritative inputs. Narrative material is chunked by semantic section at approximately 350-500 tokens with a small overlap; an outcome episode stays intact as one record. [C3]

T9 returns five candidates, filters by validation status, configuration and validity, applies a relevance threshold, reranks, and places at most three records in context. Each retains source and date. T8 assigns candidate, validated, rejected, or expired status. Only validated records are eligible for retrieval; an outcome-based lesson requires reconciliation with observed results where applicable. Similar wording alone is insufficient evidence of operational relevance. [C3,C4]

Short-term graph state preserves the current snapshot, tool results, failed checks, candidate lineage, retries, and decision status. PostgreSQL/pgvector and graph checkpoints support persistent records and external memory. Validated feedback can inform later confidence or fallback selection; raw reflection never becomes policy automatically. This applies the feedback idea associated with Reflexion through an explicit admission gate. [C2,C4; 10]

3.2 Two Agents with Deterministic Control

Role	Structured contribution	Boundary
Thought Generator	Up to three evidence-linked interpretations or documented fallback strategies from the same snapshot.	Cannot invent measurements, calculate availability, or relax constraints.
Critic / Evaluator	Scores grounding, agreement, consistency, uncertainty, and usefulness; identifies unsupported assumptions.	Ranking is provisional until hard checks pass. Its score is not a calibrated probability.
Graph / beam controller	Retains alternatives, enforces budgets, promotes a valid survivor, or abstains.	Deterministic Python and LangGraph; not a third LLM agent.

Selective Tree of Thoughts is triggered when several grounded interpretations remain plausible, including forecast disagreement or competing fallbacks. It adapts generate-evaluate-search to uncertainty interpretation, not reserve arithmetic. Retaining an alternative supports recovery when the preferred branch fails. Separating generator and critic creates distinct roles, but does not guarantee statistically independent errors. [C4,C5; 2,3]

3.3 Final Search and Acceptance Policy

Control	Consolidated policy
Search limits	Up to three candidates per expansion; beam width two; maximum depth three; at most one revision round; no more than eight total LLM calls per ambiguous episode.
Hard acceptance	T3 checks evidence before reasoning. Each candidate must pass T4/T5/T6 before final acceptance; hard failures are pruned regardless of model rank.
Rubric / stopping	Grounding 25; freshness/agreement 20; consistency 20; calibration 20; usefulness 15. Early stop requires hard validity, score at least 80/100, and a lead of at least 10 points.
Close scores / exhaustion	Within five points, prefer stronger provenance and lower uncertainty; otherwise preserve alternatives within the beam. If no supported acceptable branch survives, withhold or escalate.

The final policy applies the later one-revision limit together with the original depth and call ceilings; whichever limit is reached first stops expansion. Nested model calls and retries consume the same budget. Depth describes a reasoning stage, not permission for extra revision rounds. These are design settings, not experimentally optimized values. [C4,C5]

4. Design Evolution Across the Program

Checkpoint	Design change	Why the change mattered
1.1 - Concept	Defined the 12-24 hour PAP, operator use case, modular coordinating agent, data boundary, and repeated forecast cycle.	Established that fresh numerical evidence and equipment constraints must drive the answer.
2.1 - Reasoning	Added a ReAct-inspired loop, short/long-term memory, T1-T8 interfaces, bounded fallback, and independent validation.	Replaced prompt-only behavior with observation, tool results, revision, publication, and outcome feedback.
3.1 - Grounding	Added T9 semantic retrieval; five candidates reduce to at most three validated contextual records.	Made historical guidance searchable without substituting text retrieval for live measurements or calculations.
4.1 - Search	Selected LangGraph as workflow owner, bounded ToT, LangChain generator/critic nodes, and explicit T8 admission states.	Preserved alternatives under ambiguity while bounding search. CrewAI was excluded from the MVP to avoid duplicate orchestration.
5.1 - Coordination	Limited the system to two LLM agents on ambiguous cases, typed communication, beam width two, and one revision round.	Provided proposal/critique separation and branch recovery without open-ended agent conversation.
6.1 - Assurance	Added layered input/runtime/output guardrails, multi-metric evaluation, logging, calibration monitoring, and risk-based human review.	Made unresolved uncertainty change publication behavior, with explicit withholding and escalation.

Across all six checkpoints, the stable principle is that the model interprets evidence while deterministic services enforce numerical and policy constraints. ReAct motivates interleaving reasoning with observations; ToT adds selective search; the later checkpoints narrow agent autonomy rather than granting it equipment authority. [C1-C6; 1,2]

Appendix A (Figures 2-7) shows the inputs, decisions, outputs, and recovery paths introduced by each of Modules 1-6.

5. Implementation Overview

The implementation is organized as a Python workflow with typed graph state and modular T1-T9 interfaces. The development sequence establishes the deterministic PAP path and persistence first, then publication and outcome evaluation, then validated pgvector retrieval, selective ToT, and bounded LangChain generator/critic nodes. Local PostgreSQL is the intended prototype data store. The final runtime architecture is specified in Checkpoints 4.1-6.1.

Prior development updates report staged pull requests completed through PR11. That is reported implementation progress; the six reviewed PDFs do not contain the repository snapshot, dependency lockfile, model configuration, executable test output, or deployment evidence. The descriptions here consolidate the final implementation design and reported progress without treating those updates as measured benchmark results.

Data contract	Minimum content
Run / evidence state	Run identifier; forecast horizon; coherent timestamps; normalized inputs; configuration and policy versions; source and evidence identifiers.
Candidate / decision	Evidence-linked strategy; objections and rubric score; numerical result reference; hard-check outcomes; branch lineage; budget counters; decision status.
Published PAP	Interval start/end and available kW; uncertainty or confidence basis; freshness; provenance; applicable limits; guidance; publication status and trace reference.

5.1 End-to-End Execution and Numerical Boundary

A cycle captures the operator horizon and approved policy, reads T1/T2, and normalizes with T3. T9 adds eligible context when useful. The graph chooses the routine strategy or the bounded ambiguous path, then invokes T4 for interval forecasts and T5 for availability. T6 independently checks the candidate. The publication gate validates the final schema, evidence coverage, and risk status; T7 publishes or the graph records an explicit insufficient-evidence result. T8 later reconciles forecasts with outcomes. [C2-C6]

The numerical contract requires a coherent energy trajectory across the horizon: interval power in kW is integrated over duration in hours to update energy in kWh; approved capacity, losses, reserve, and power limits constrain that trajectory. A profile must preserve the reserve floor at every interval. Detailed equipment parameters and a complete formula implementation are not supplied in the checkpoints, so this report does not invent battery ratings, efficiencies, reserve percentages, or forecast-error tolerances.

Retrieval may lower confidence or recommend re-evaluation while retaining the same calculated availability, as in Checkpoint 3.1. If a later interpretation selects a different approved forecast or fallback, its numerical inputs must be recorded and T4/T5/T6 rerun. Neither retrieved prose nor the Critic may directly edit the validated power values. [C3,C4]

6. Evaluation Methods and Results

6.1 Evaluation Method

The evaluation design separates numerical forecasting, retrieval quality, agent decisions, and operational reliability. It uses held-out synthetic or anonymized episodes covering normal conditions, variable clouds, low SOC, high wind, source disagreement, stale or malformed data, configuration mismatch, and branch failure. HELM informs the choice to assess multiple scenarios and metrics rather than a single accuracy score. [C3-C6; 5]

A reproducible comparison should run the linear baseline, baseline plus RAG, and selective two-agent ToT on the same evidence snapshots. Forecast models, numerical constraints, and approved inputs remain fixed except for the feature under test. Split episodes by time and configuration where feasible; exclude test outcomes from retrieval and calibration. Record corpus, policy, prompt, model, and code versions, and repeat probabilistic runs to expose variability. These are the final evaluation protocol, not a claim that the benchmark has been executed.

Metric	Operational definition / required reporting
Forecast / PAP error	Compare predicted interval power with held-out reference outcomes; report MAE in kW and energy error in kWh, with overestimation separated from underestimation.
Hard-constraint escape	Published invalid profiles divided by all published profiles, with numerator and denominator. Target: zero observed escapes; also report publication coverage.
Grounding / retrieval	Evidence-dependent claims supported by eligible sources divided by such claims; separately inspect irrelevant, expired, and configuration-mismatched retrieval.
Calibration	Define a correctness event and tolerance before computing ECE. Bin calibrated forecast probabilities and compare predicted confidence with observed correctness.
Guardrail performance	Blocked known-bad cases divided by known-bad cases; also measure valid cases blocked, by failure type.
Recovery / oversight	Valid recoveries divided by eligible recovery cases; escalations per run; reviewer changes or rejections per reviewed PAP.
Latency / cost	End-to-end p50/p95 latency, calls, tokens, and cost, split by routine and ambiguous paths; include abstentions and failures.

6.2 Results Supported by the Checkpoints

The checkpoint record contains two qualitative scenario traces. They illustrate the intended observable behavior and give concrete evaluation cases. Neither includes raw execution logs, dataset size, repeated trials, or numerical performance measurements. They are reported checkpoint evidence, not newly executed experiments or a demonstrated field-performance improvement. [C3,C5]

Reported case	Behavior documented	Supported conclusion
Retrieval fixture Checkpoint 3.1	Under changing cloud cover, low SOC, and elevated wind, five candidates are filtered/reranked to three validated records. Retrieved episodes lower confidence and add a recommendation to re-evaluate before optional load is committed. Calculated availability stays unchanged.	Context can affect interpretation and operator guidance while the physical calculation retains authority. No numerical calibration gain or retrieval accuracy rate is reported.
Beam recovery trace Checkpoint 5.1	Forecast-source disagreement produces A and B. The Critic initially prefers A. T6 rejects A for violating the reserve floor. Beam width two preserves B; B passes hard checks and is promoted.	The documented recovery logic preserves an alternative and allows deterministic validation to reject model preference. No recovery success rate is reported.

6.3 Interpretation of Results and Remaining Evidence

The strongest supported project result is an integrated design with explicit evidence, calculation, search, publication, and review boundaries. The retrieval fixture and recovery trace explain how those boundaries should change behavior. They do not establish that selective ToT improves forecast accuracy, that the Critic is calibrated, or that the full system meets a production safety or latency target.

Evidence category	Current status in the reviewed record
Architecture and design evolution	Documented across all six checkpoints, including tool roles, retrieval policy, search limits, guardrails, and oversight.
Qualitative behavior	One synthetic retrieval fixture and one ambiguous-case recovery trace are described. Their evidence is limited to the reported examples.
Implementation progress	Prior project updates report completion through PR11. A code snapshot, exact runtime versions, and independently inspected test outputs are absent from this review.
Quantitative results	No measured PAP error, ECE, escape rate, grounding rate, detection rate, recovery rate, escalation rate, or latency/cost distribution is supplied.
Production readiness	No field validation, deployment acceptance, certified equipment control, or independently verified safety guarantee is claimed.

The prototype target of zero hard-constraint escapes is an acceptance goal, not an observed result. A benchmark should also report how often valid profiles are published, because withholding every result can produce zero escapes without providing useful service. Small or empty denominators must remain visible; a run with no published profiles has no measurable published-profile escape rate. [C6]

Before reporting empirical confidence, define what a forecast probability means: for example, that availability remains within an approved error tolerance while preserving all modeled constraints. The tolerance must be set with the operator before testing. A rubric score of 80/100 is a branch-ranking threshold, not an 80% probability of correctness. Model self-evaluation can be useful but can miscalibrate on new tasks and conditions. [C4,C6; 6]

7. Safety, Reliability, and Human Oversight

The course system is read-only, but a misleading forecast can still influence a consequential operator decision. Safety therefore depends on layered controls and transparent withholding, as well as the absence of control tools. Confidence is advisory; hard numerical limits and evidence requirements determine whether a PAP may be published. [C6]

Layer	Control	Response to failure
Input	T3 checks completeness, units, plausible ranges, freshness, and coherent timestamps. T9 uses validated, relevant, in-date records.	Refresh approved sources; discard ineligible context; stop before reasoning if critical data remain unavailable.
Runtime	Read-only tool allowlist; bounded branches, depth, revision, retries, and total model calls. Treat retrieved content as evidence rather than authority to change policy.	Use an approved fallback within budget; stop or escalate when evidence or budget is exhausted.
Numerical	T5 accounts for interval energy and reserve. T6 enforces configured equipment, reserve, freshness, and output invariants.	Prune invalid candidates; promote a preserved valid alternative; withhold if none survives.
Publication	Validate typed fields, supported explanations, provenance, uncertainty basis, constraints, risk status, and trace completeness.	Withhold the PAP or issue an explicit insufficient-evidence status; unresolved high risk goes to human review.
Feedback	Compare with outcomes; monitor bias, calibration, drift, and overrides; apply T8 admission and expiry.	Tighten routing or confidence policy through a validated change; prevent raw feedback from silently changing limits.

7.1 Risk-Based Human Intervention

Risk	Autonomous behavior	Human role
Low	Fresh evidence, hard checks passed, supported guidance, and uncertainty within a calibrated operating range.	No routine review; publish the read-only PAP and retain the trace.
Elevated	Ambiguity triggers retrieval, critique, or bounded fallback/ToT.	Review when automated recovery cannot establish a valid, supported result.
High	Critical data missing; no valid branch; unresolved source conflict; confidence below a calibrated threshold; policy/configuration unknown; or drift alarm.	Mandatory review. PAP remains withheld or insufficient until the evidence, policy, or configuration problem is resolved.

The reviewer receives a coherent evidence snapshot, failed checks, candidate comparison, and the reason for escalation. Available decisions are to accept a valid low-risk PAP, request refresh/re-run, or reject/escalate. Review cannot waive equipment limits or convert an invalid candidate into a published valid profile. A policy change must be authorized, versioned, and followed by a fresh calculation and validation. Human review grants no equipment-control capability. [C6]

7.2 Observability and Controlled Improvement

Each run records evidence identifiers and freshness, tool calls and latency, candidate scores and objections, hard failures, branch promotions, fallbacks, final status, and human decisions. These are operational decision records and concise evidence-linked rationales. Dashboards should separate routine, ambiguous, withheld, and reviewed cases so aggregate success does not hide weak operating regimes. [C6]

Operator feedback and observed outcomes inform prompts, routing, calibration, and the admitted corpus. This is controlled application-level improvement, not a claim of training the model with RLHF. The course lesson from feedback-based alignment is applied by retaining independent evaluation and checking whether the chosen proxy actually tracks useful, reliable behavior. [C3-C6; 4]

8. Current Limitations and Future Improvements

Limitation	Practical implication	Next improvement
Incomplete empirical evidence	The checkpoint traces do not establish average performance, rare-failure behavior, cost, or field reliability.	Freeze a reproducible build and run the held-out comparison described in Section 6; publish raw counts and per-scenario metrics.
Unverified calibration	Numerical confidence cannot be treated as an empirical probability without sufficient labeled outcomes, especially under changing conditions.	Define correctness/tolerance, fit calibration on separate data, monitor by regime, and route unsupported conditions to review.
Limited data and physical modeling detail	Public/synthetic/anonymized inputs may not represent every device configuration, battery condition, or weather regime. Equipment parameters are not supplied in the checkpoint package.	Validate adapters and energy accounting against approved device data; add representative anonymized episodes and configuration-specific independent checks.
Retrieval and correlated model errors	A plausible memory or two agreeing agents can share an incorrect assumption. Beam pruning can discard a better interpretation.	Test expired/mismatched/adversarial context, evaluate retrieval separately, preserve alternatives, and compare critic decisions with deterministic evidence and reviewer judgments.
Operational overhead	Search, critique, persistence, and escalation add latency and can overload reviewers if thresholds are too strict.	Measure costs by route, keep routine cases simple, and tune budgets and review thresholds only against held-out reliability and workload results.
Read-only course boundary	The project does not demonstrate autonomous physical control or justify using a forecast as an equipment interlock.	Begin with operator decision support and shadow evaluation. Treat any later control capability as a separate project with independent engineering validation.

8.1 Prioritized Next Steps

First, make the evidence reproducible. Capture the current repository revision, locked dependencies, model and prompt settings, policy/configuration versions, corpus snapshot, and test manifest. Link each claimed behavior to a repeatable run. This closes the gap between reported implementation progress and inspectable implementation evidence.

Second, evaluate the smallest complete workflow. Establish the deterministic baseline, then measure the incremental contribution of RAG and selective ToT on identical snapshots. Keep the additional agents only where they improve supported decisions or recovery enough to justify their cost. Guardrail testing must include false positives and failure cases, not just successful publication.

Third, validate with operators. Use read-only shadow runs, compare forecasts with outcomes, assess whether explanations help load-planning decisions, and use review outcomes to improve calibration and routing. Expand to more sites or data sources only after the same policy, versioning, and observability boundaries remain intact.

8.2 Program Reflection

The central learning from the capstone is that reliable autonomy comes from assigning authority carefully. External data determines what the system knows; numerical tools determine what is feasible; retrieval adds qualified context; model roles compare interpretations; and the graph and human oversight determine whether to publish or stop. Each checkpoint introduced a capability together with a boundary that makes its failure observable.

The final contribution is a coherent PAP decision-support design that combines grounding, selective search, two-agent critique, deterministic validation, and controlled feedback. The documented example traces make its intended behavior concrete. The next milestone is measured validation of those behaviors on a reproducible implementation, followed by operator evaluation under the same read-only boundary.

References and Checkpoint Source Record

Project claims in this report are traced to all six supplied checkpoint PDFs, totaling 14 pages. The checkpoint references identify design and reported scenario evidence; external references establish the research and framework background. Prior development updates are identified as reported progress where used.

Capstone Checkpoints

- [C1] Shawn Cunningham. *Required Capstone Checkpoint 1.1: Power Availability Profile Forecaster Agent*. Initial concept; 3 pages. File: Capstone_Checkpoint_1_1_DragonWings_PAP_Forecaster.pdf.
- [C2] Shawn Cunningham. *A ReAct-Inspired Architecture for a Power Availability Profile Forecaster Agent*. Checkpoint 2.1, Section B, August 2026; 3 pages. File: 20260802_Capstone_Checkpoint_2_1_PAP.pdf.
- [C3] Shawn Cunningham. *A Retrieval-Grounded Architecture for a Power Availability Profile Forecaster Agent*. Checkpoint 3.1, Section B, August 2026; 2 pages. File: 20260810_Capstone_Checkpoint_3_1_PAP_Section_B_FINAL (1).pdf.
- [C4] Shawn Cunningham. *A Selective Tree-of-Thought Architecture for a Power Availability Profile Forecaster Agent*. Checkpoint 4.1, Section B, August 2026; 2 pages. File: 20260816_Capstone_Checkpoint_4_1_PAP_Section_B_FINAL.pdf.
- [C5] Shawn Cunningham. *A Minimal Graph-Coordinated Multi-Agent Architecture for a Power Availability Profile Forecaster Agent*. Checkpoint 5.1, Section B, August 2026; 2 pages. File: 20260823_Capstone_Checkpoint_5_1_PAP_Section_B_FINAL_REVISIED.pdf.
- [C6] Shawn Cunningham. *A Layered Safety and Human Intervention Plan for a Power Availability Profile Forecaster Agent*. Checkpoint 6.1, Section B, September 2026; 2 pages. File: 20260901_Capstone_Checkpoint_6_1_PAP_Section_B_FINAL.pdf.

Research and Framework Sources

- [1] S. Yao, J. Zhao, D. Yu, et al. *ReAct: Synergizing Reasoning and Acting in Language Models*. arXiv:2210.03629, 2022; ICLR 2023.
- [2] S. Yao, D. Yu, J. Zhao, et al. *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*. NeurIPS 2023; arXiv:2305.10601.
- [3] T. Guo, X. Chen, Y. Wang, et al. *Large Language Model based Multi-Agents: A Survey of Progress and Challenges*. arXiv:2402.01680, 2024.
- [4] Y. Bai, A. Jones, K. Ndousse, et al. *Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback*. arXiv:2204.05862, 2022.
- [5] P. Liang, R. Bommasani, T. Lee, et al. *Holistic Evaluation of Language Models*. TMLR, 2023; arXiv:2211.09110.
- [6] S. Kadavath, T. Conerly, A. Askell, et al. *Language Models (Mostly) Know What They Know*. arXiv:2207.05221, 2022.
- [7] LangChain. *LangGraph Overview*. Official documentation; accessed September 7, 2026.
- [8] LangChain. *Agents*. Official documentation, create_agent; accessed September 7, 2026.
- [9] Model Context Protocol. *Architecture Overview*. Official documentation; accessed September 7, 2026.
- [10] N. Shinn, F. Cassano, E. Berman, et al. *Reflexion: Language Agents with Verbal Reinforcement Learning*. arXiv:2303.11366, 2023.

Appendix A. Module Workflows

Figures 2-7 show how Modules 1-6 contribute to the PAP workflow. Red outlines mark decisions, hard checks, and withholding; gray fills mark stored data. The charts describe the consolidated design and reported examples.

A.1 Module 1 - The Forecasting Task

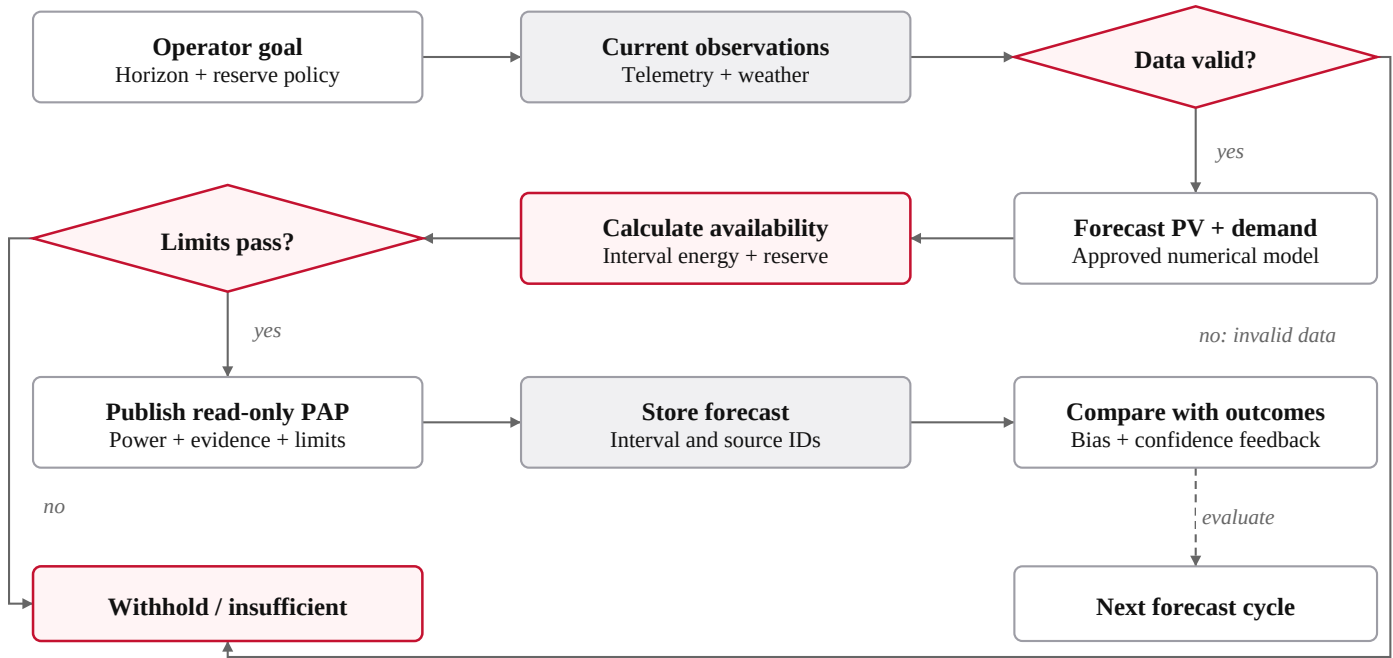


Figure 2. A forecast cycle begins with the operator goal and current evidence, applies numerical and reserve checks, and publishes a read-only profile. Invalid evidence or limits produce an explicit insufficient result. Outcome comparison informs the next cycle. [C1]

A.2 Module 2 - ReAct and Memory

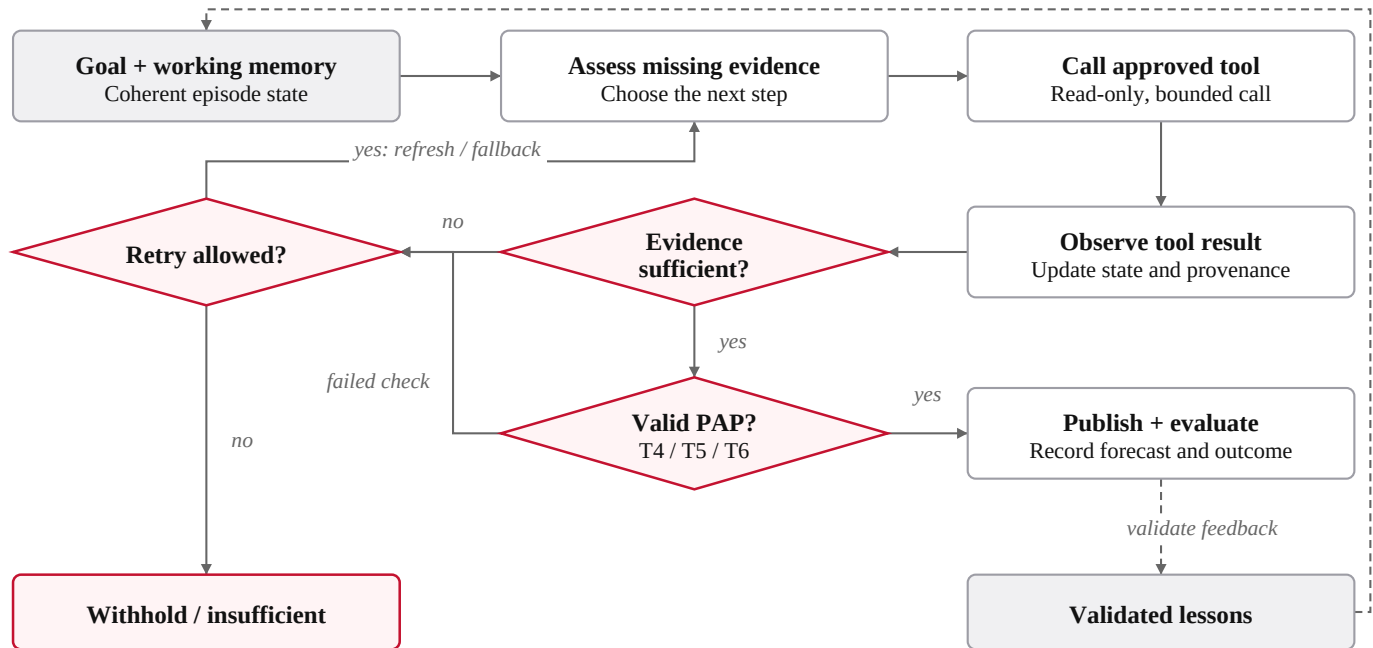


Figure 3. Tool results update the next assessment. Refresh and fallback remain bounded; numerical failures cannot be published. Short-term state supports the current episode, while validated lessons support later cycles. Dashed arrows show evaluation or memory feedback. [C2]

Appendix A. Module Workflows (continued)

A.3 Module 3 - Retrieval and Grounding

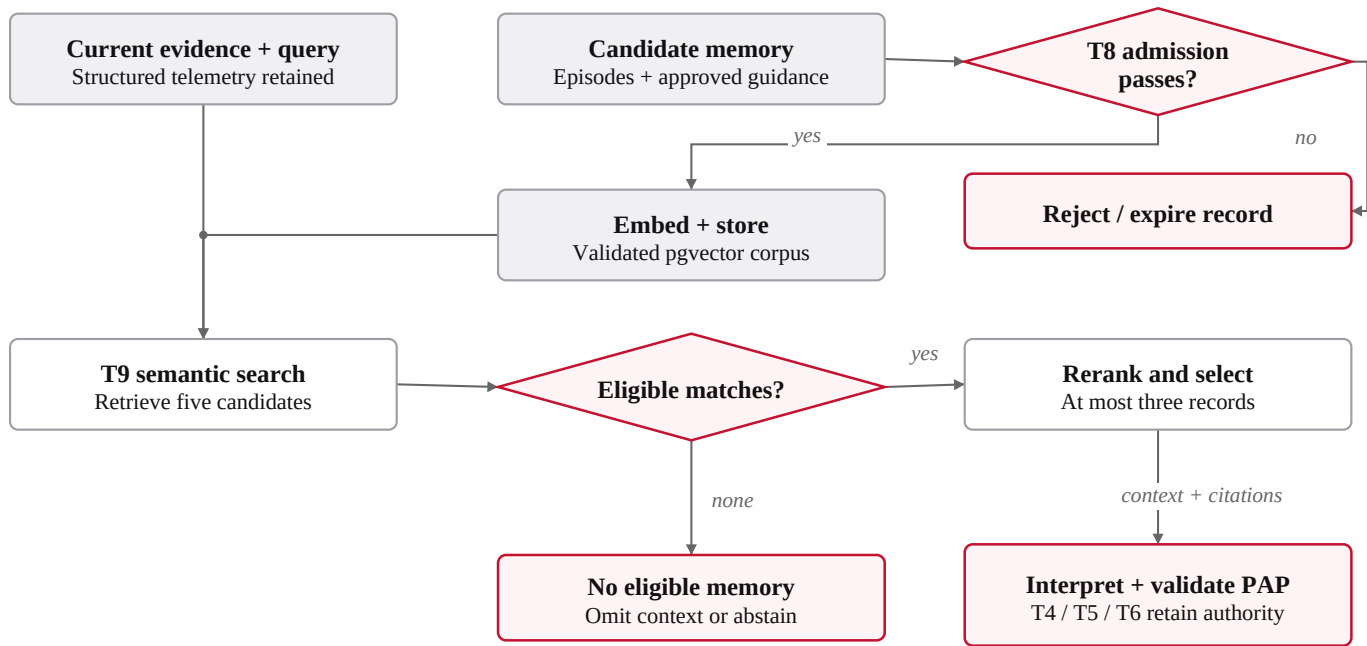


Figure 4. T8 controls admission to memory; T9 retrieves five candidates and retains at most three after configuration, validity, relevance, and provenance checks. Context can affect explanation and confidence; numerical calculation and validation retain authority. [C3,C4]

A.4 Module 4 - Selective Tree of Thoughts

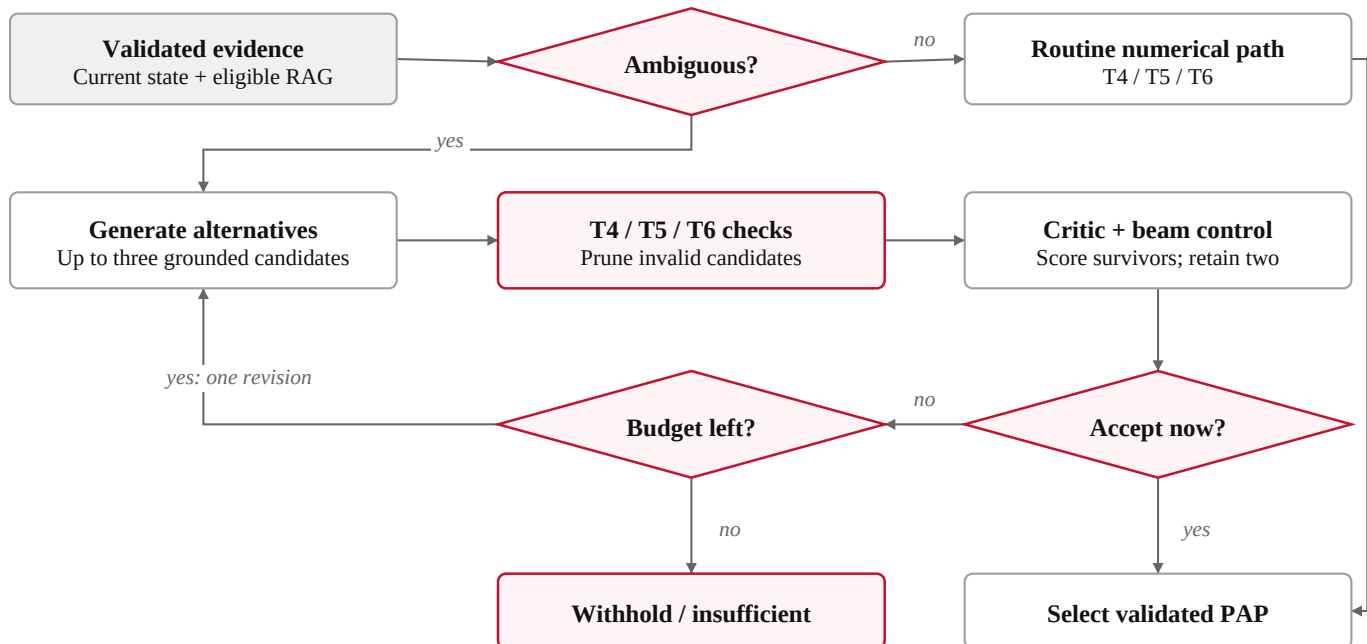


Figure 5. Search is limited to depth three, beam width two, one revision round, and eight LLM calls. Early acceptance requires hard validity, score at least 80, and a lead of at least 10 points. At a stopping boundary, select the best supported acceptable survivor; otherwise withhold. [C4,C5]

Appendix A. Module Workflows (continued)

A.5 Module 5 - Two-Agent Coordination and Recovery

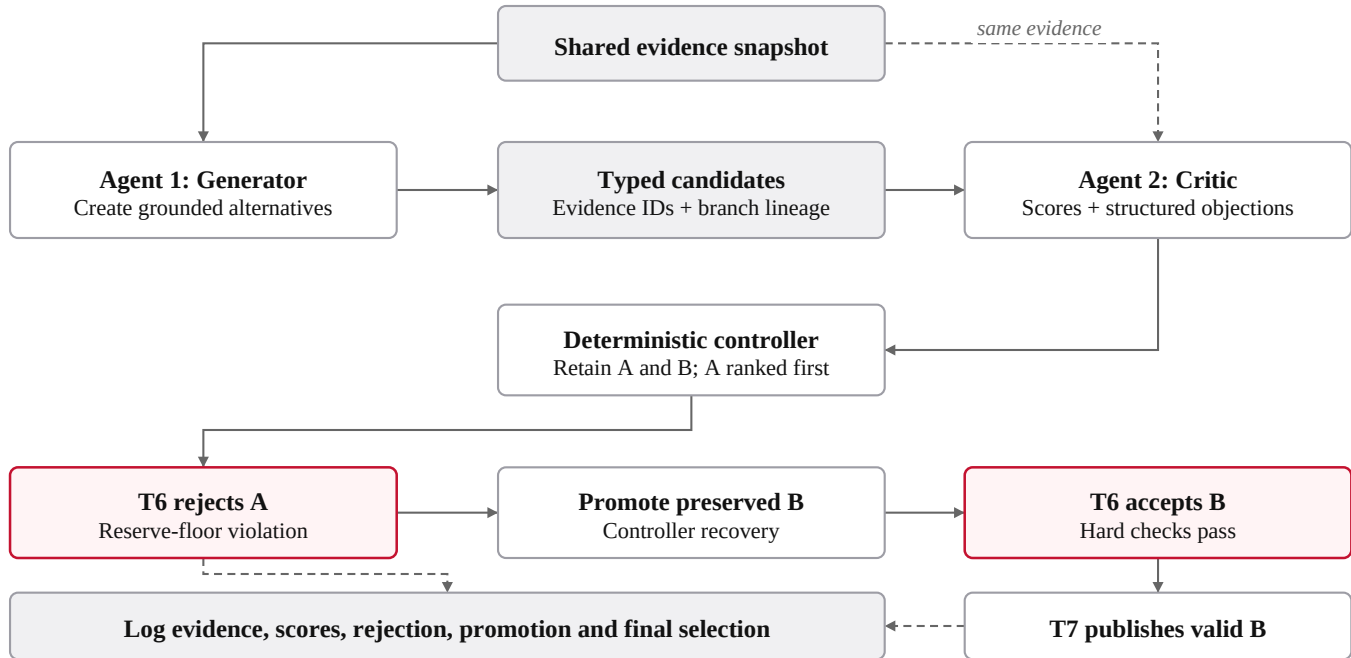


Figure 6. The reported recovery trace: generator and critic share evidence, but the deterministic controller owns selection. A reserve violation rejects preferred A; preserved B is validated and published. This is a qualitative checkpoint example, not a measured recovery rate. [C5]

A.6 Module 6 - Safety, Reliability and Human Review

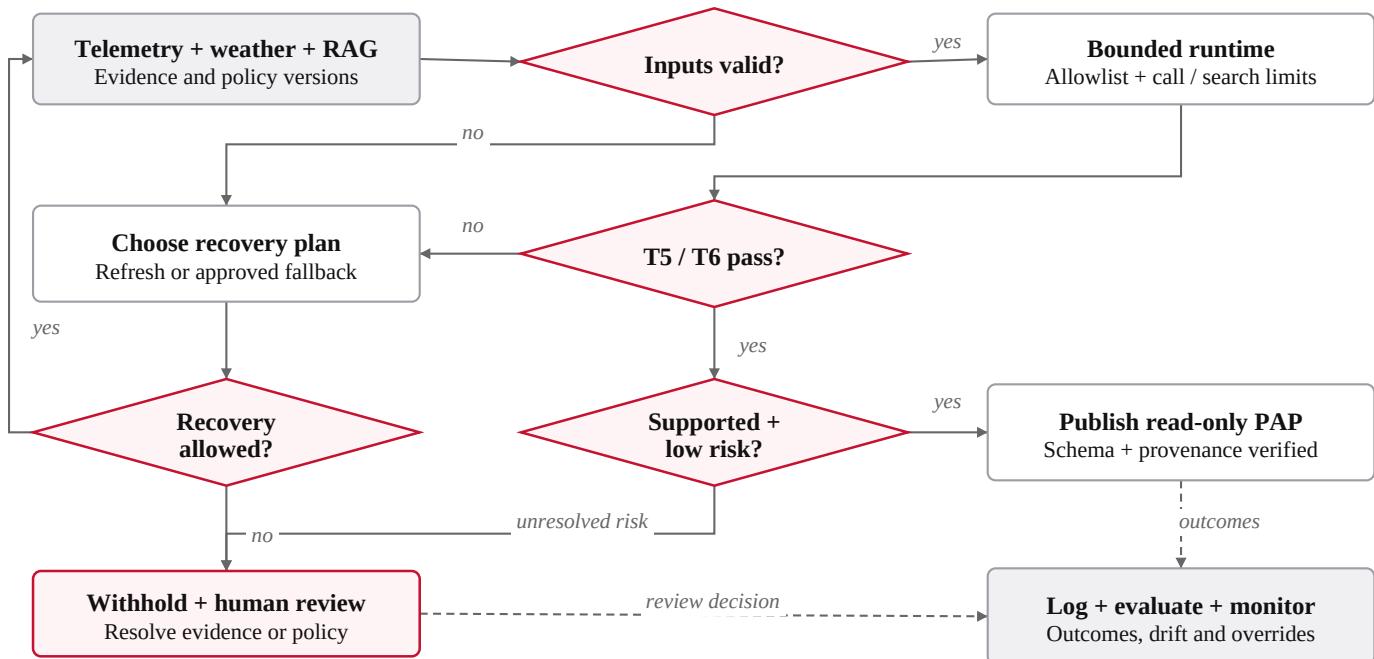


Figure 7. Input, runtime, numerical, and publication checks bound autonomy. Failed recovery or unresolved risk withholds the PAP and triggers review. A reviewer resolves evidence or policy and requests a new validated run; review cannot bypass hard checks. [C6]